Making sense of data

CE/EL/MC/CV/GL/GM/ES 361 Probability and Statistics • First Semester, 2026/2027

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Week 1 • Chapter 1, Introduction to Statistics

How to use this sheet

Try every question before the tutorial, in your group if you like, and bring your working on paper. Several of these questions have no single correct answer: they ask you to take a position and defend it with the numbers. A one line answer with no reasoning earns nothing, and a wrong answer argued carefully is worth more than a right answer copied from a friend.

Q1. A mine logs the tyre pressure of every haul truck tyre checked during one month, 9 400 checks in all. An engineer takes every 20th record from the log, giving 470 readings with a mean of 112 psi and a standard deviation of 9 psi.

(a) Name the population, the sample, the parameter and the statistic in this study.

(b) Is 112 psi a parameter or a statistic? Is there a parameter anywhere in the question that you can actually see?

(c) The engineer writes in her report: "the fleet mean tyre pressure is 112 psi." What is missing from that sentence? Rewrite it as a statistician would.

(d) Taking every 20th record is cheap and feels random. Describe one realistic way the mine's record keeping could make it badly wrong.

Q2. Classify each of the following as nominal, ordinal, interval or ratio, and in one line say what gave it away.

Variable	Recorded as	Variable	Recorded as
Blast hole identifier	BH-104	Ore type	oxide, sulphide
Mohs hardness rank	1 to 10	Shift	A, B, C
Temperature of the mill bearing	°C	Time to failure	hours
Operator rating of a machine	1 to 5	Easting of a survey peg	metres

(a) A colleague reports that "the mean Mohs hardness of our samples is 5.4". Why should that sentence make you uncomfortable, even though you can certainly compute it?

(b) You can say a 40 kg sample is twice as heavy as a 20 kg one, but you cannot say 40°C is twice as hot as 20°C. Explain the difference, and say which property of the scale is responsible.

Q3. You must estimate the average daily turbidity of a river downstream of a mine over one year. Four designs are proposed.

(a) Sample every Monday morning at 08:00.

(b) Sample whenever the plant manager says conditions are "normal".

(c) Sample on 52 days chosen at random from the year.

(d) Divide the year into wet and dry season, and sample at random within each.

Rank the four from most to least trustworthy. For each one, name the bias it invites and say in which direction you expect it to push the estimate. Then answer this: you are given the budget for only 12 visits in the year. Which design do you use, and what do you lose?

Q4. Fifteen turbidity readings (NTU) were recorded at the same sampling point:

12, 14, 15, 15, 16, 17, 18, 19, 21, 22, 24, 27, 31, 38, 46

(a) Find the mean, median, mode, range, the quartiles, the interquartile range and the standard deviation.

(b) Apply the $1.5 \times$ IQR rule. Which reading, if any, is flagged?

(c) The field book is checked and the last reading was a transcription error: it should be 4.6, not 46. Recompute all the summaries in (a).

(d) Which summaries moved a lot and which hardly moved? Say in one sentence what that tells you about choosing a summary when your data may contain recording errors.

(e) The regulator asks for a single number describing the river that month. Which one do you give, and what do you say alongside it?

Q5. A maintenance team records the time to repair a conveyor fault, in hours, on fifteen occasions:

$$0.5,\ 0.7,\ 0.8,\ 1.0,\ 1.0,\ 1.2,\ 1.5,\ 1.8,\ 2.0,\ 2.4,\ 3.1,\ 4.0,\ 6.5,\ 9.0,\ 14.0$$

(a) Compute the mean, the median and the mode, and sketch the shape of the data.

(b) The mean is nearly twice the median. Which way is the distribution skewed, and which observations are responsible?

(c) The maintenance manager wants to report "typical repair time" to his director. Which average will he prefer, and is he being dishonest?

(d) The spare parts planner needs to know how many repair hours to budget for the next 60 faults. Which average must he use, and why is it the other one?

Q6. A clerk summarises 50 blasting fragment sizes into classes and draws a bar chart of the frequencies.

Class (cm)	0–10	10–20	20–30	30–40	40–60
Frequency	4	11	18	9	8

(a) Why is a bar chart of these frequencies misleading?

(b) Compute the frequency density for each class and say which class is really the busiest, and which the chart makes look busier than it is.

(c) Estimate the mean fragment size from the table. You cannot get the exact mean from it: explain precisely what information was lost, and what you would have to keep to avoid the loss.

Q7. Three processes are monitored over the same shift.

	Mean	Standard deviation
Crusher A throughput (t/h)	420	35
Crusher B throughput (t/h)	180	21
Mill feed grade (g/t)	2.4	0.6

(a) Which process has the largest standard deviation, and which is the most variable? Explain why these are not the same question.

(b) Compute the coefficient of variation for each.

(c) A supervisor says the grade is "the most stable of the three, look at how small its standard deviation is". Write the two sentence reply you would give him.

Q8. Criticise each of the following, and say how you would redraw or rewrite it.

(a) A bar chart of plant availability for four months, with the vertical axis running from 95% to 100%, shown to the board as evidence of a "dramatic improvement".

(b) A pie chart of accident causes with eleven slices, whose percentages add up to 103%.

(c) A line joining the mean strength of concrete at 7, 14 and 28 days, extended by the same line to predict the strength at 365 days.

(d) Is a truncated vertical axis always wrong? Give a case where it is the right choice.

Q9. From the table in Q6, state the modal class and estimate the median class. Then answer: if you were told the raw 50 measurements had a mean of 25.8 cm, by how much did the grouped estimate miss, and is a miss of that size a problem for a blasting engineer? Defend your answer.

Q10. Say whether each statement is always true, sometimes true, or never true, and justify it in one or two lines.

(a) "The mean density of our rock samples is 2.7 g/cm^3, so half of the samples are denser than 2.7."

(b) "No student in the class scored above the class average."

(c) "Adding one more observation cannot change the median by more than it changes the mean."

(d) "If two datasets have the same mean and the same standard deviation, they have the same shape."

The datasets here are made up for teaching, but the situations are not. Bring your working to the tutorial: the discussion is the point of the sheet, not the arithmetic.